Maximize Expression

AlgoExpert · Hard · Dynamic Programming

The problem

- Max of a - b + c - d over indices a < b < c < d (0 if fewer than 4 elements).
- Four rolling DP layers.

Intuition

- Build the expression one term at a time: best a, then best a minus b, and so on, each over a prefix.

Approach

- 1. Four arrays of running bests
- 2. Each layer uses the previous layer up to the index before
- 3. Return the last layer's max

Java solution

```java
public class MaximizeExpression {
    public static int maximizeExpression(int[] a) {
        if (a.length < 4) return 0;
        long s1 = Long.MIN_VALUE / 4, s2 = Long.MIN_VALUE / 4, s3 = Long.MIN_VALUE / 4, s4 = Long.MIN_VALUE / 4;
        for (int i = 0; i < a.length; i++) {
            if (i >= 3) s4 = Math.max(s4, s3 - a[i]);
            if (i >= 2) s3 = Math.max(s3, s2 + a[i]);
            if (i >= 1) s2 = Math.max(s2, s1 - a[i]);
            s1 = Math.max(s1, a[i]);
        }
        return (int) s4;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Build the expression one term at a time: best a, then best a minus b, and so on, each over a prefix.
- Complexity: O(n)